

PROJECT DESIGN

POCKETSMART AI — Smart Budget & Recommendation Assistant

Expanded phase evidence with detailed wording, diagrams, workflow, and project-focused explanation.

Design objective

The design phase converts requirements into a layered application structure. PocketSmart AI separates the user interface, FastAPI routing, planner-specific processing, Gemini integration, authentication, data models, and History. This separation makes the workflow easier to understand and provides a foundation for maintaining the project as additional planning categories or richer recommendation logic are introduced.

System architecture

At the centre of the design is a consistent request-to-result pipeline: User Input → Validation → FastAPI Route → Planner Logic → Gemini API → Result Formatting → Template Display → History. The architecture keeps domain-specific inputs inside the corresponding planner while allowing the overall application to reuse common authentication, routing, result, and History concepts.

Interface and data design

The frontend is based on Jinja2 templates with HTML/CSS. The backend uses Pydantic models for structured planner inputs. Authentication uses password hashing and JWT/access-token handling through HTTP-only cookies. The Gemini utility layer keeps AI interaction separate from page rendering. This design gives each responsibility a recognizable location in the project structure.

Visual Model



Figure — Phase visual model

Process Flow



Figure — Detailed workflow / roadmap

PHASE 3 — DETAILED CONTENT

PocketSmart AI connects practical planning with a structured recommendation workflow. The following section expands the phase in clear project language so the document has sufficient explanatory depth for academic/project review.

Project Explanation

A strong budget-planning assistant should reduce uncertainty without hiding the reasoning context behind the result. In PocketSmart AI, the user supplies the information that matters for the selected category, including budget and relevant preferences. The application then processes that context and presents a structured recommendation. This structure makes the output easier to read, compare, and explain during a project demonstration. The project also demonstrates how a common architecture can support different use cases. Home Interior planning focuses on rooms, furniture, lighting, fans, and dining requirements. Party planning focuses on guests, event type, venue type, catering, decoration, and entertainment. Jewelry planning focuses on occasion, style preferences, budget, and optional outfit-image context. Although the inputs differ, the overall application pattern remains consistent.

Key Concepts

Layer	Responsibility
Frontend / Templates	Forms, navigation, cards, summaries, recommendation pages.
FastAPI Backend	Routing, validation, authentication, application coordination.
Planner Modules	Home, Party, and Jewelry-specific requirement processing.
Gemini Integration	Context-aware recommendation generation.
History	Display and continuity for previous recommendation results.

Roadmap / Mind-Map View

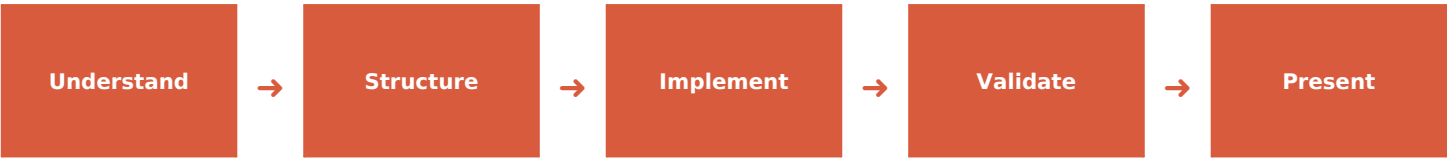


Figure — Additional project roadmap visual

Project Evidence

The supplied project documentation includes interface and workflow evidence. A representative screenshot is included here to connect the written explanation with the implemented application.

PocketSmart AI — End-to-End Application Workflow

From secure access to budget-aware, Gemini-powered recommendations

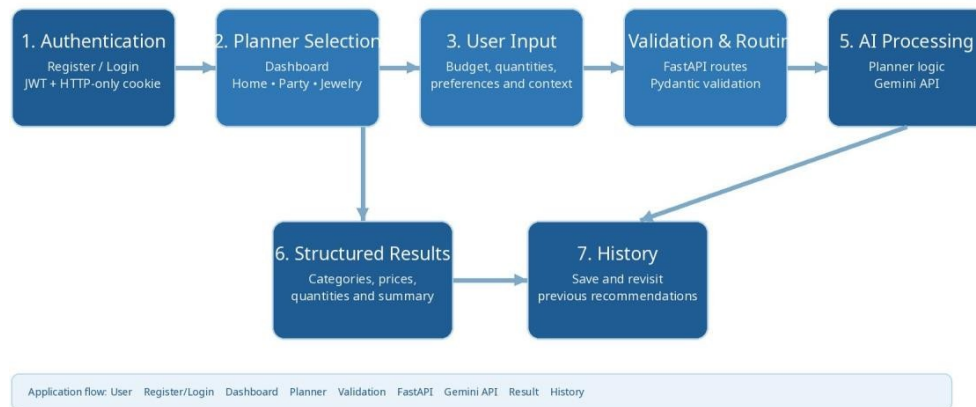
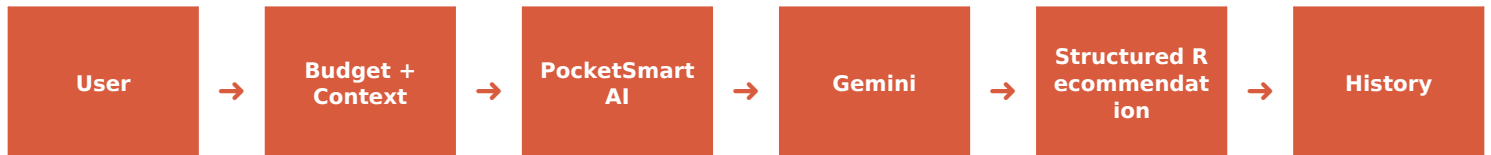


Figure — Representative PocketSmart AI project evidence

PHASE SUMMARY

Phase 3 contributes to the overall PocketSmart AI lifecycle by making the project easier to understand, build, validate, or present. The emphasis in this phase is on clear communication: the reader should be able to see what the system is doing, why the step matters, and how it connects to the final budget-aware recommendation workflow.

End-to-End Connection



Project Reflection

PocketSmart AI is designed around a practical user problem: making planning decisions while respecting a defined budget. The project brings together web development, authentication, structured data, Generative AI, visual presentation, and History. This combination makes the application a useful example of how a software project can move from a simple idea to a demonstrable end-to-end system. The documentation should therefore communicate not only the features, but also the reasoning, workflow, and evidence behind them.

Closing Note

The phase is complete when its purpose, workflow, supporting evidence, and connection to PocketSmart AI are understandable to a reviewer without requiring them to infer missing steps. The expanded paragraphs and diagrams in this version are intended to make that explanation clearer and more professional.